Danmarks Tekniske Universitet

DTU

FEM for Vibrations and Acoustics Report 1

Authors of Group 4

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Group Responsibilities

Author	Task
Laurenz Walter Torni – s252851	Problem Statement and Energies of the system
Jente Staelens – s252864	Forced Vibration

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1 Problem Statement

The present study investigates the dynamic response of a snowboard during a carving turn, with a focus on how a damping element influences its vibrational behavior and energy distribution within the board. A two-dimensional Mindlin plate finite element model is used to investigate how snow interaction, heterogeneous material composition, and rider load affect a snowboard's natural frequencies, mode shapes, and modal energy distribution. The model incorporates a central damping region embedded within the snowboard structure. The geometric setup, boundary conditions, and loading scenario are illustrated in figure 1 and chosen to realistically represent the engaged edge during a carving turn, while the material parameters used in the simulation are summarized in table 1.

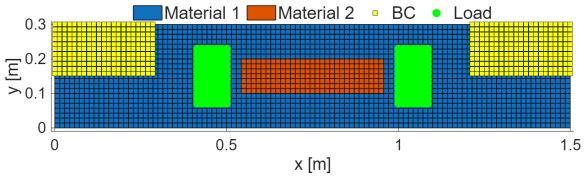


Figure 1: Used FEM-Model with a meshsize of ny = 21 and nx = 105 elements.

Table 1: Material properties.

Mat.	ρ [kg/m^3]	E [Pa]	ν	α	β
1	930	9×10^8	0.3	0.0	0.0
2	1150	5×10^7	0.3	0.5	0.0002

The snowboard has a total length $L = 1.5$ m and a nominal width $w = 0.3$ m, the thickness is chosen 100 times smaller than the width to satisfy the mindlin plate theory. The mesh consists of two materials one as the body material (material 1) and the other one as a damping component (material 2) in the center of the board. During a carving turn, the inside edge at the nose and tail engages with the snow. This interaction is modeled as two rectangular support regions at each end of the board, constraining the out-of-plane displacement (z-direction) over a length of $L_{\text{BC}} = 0.3$ m across half the board width on the inner board-edge of the carving turn. A total downward force $F = 1$ kN is applied as a uniform distributed load over two rectangular binding areas placed 0.3 m from the longitudinal center of the board. After having conducted a mesh refinement study a domain discretization with 105 elements longitudinally in x-direction and 21 transversely in y-direction (in total 2205 elements and 2332 nodes) has been chosen.

2 Forced Vibration

In Figure 2 the frequency response function (FRF) for 10 eigenmodes in a node right beneath the feet is shown for several solution methods: the direct solver, modal superposition, modal acceleration, and the Craig–Bampton direct solver. In this case, non-proportional damping is introduced due to the presence of two different material damping factors. Consequently, modal superposition must be performed using complex modal analysis, which requires expanding the eigenvalue problem to twice its original size. The theory for this implementation was found in [1]. For both the modal superposition and modal acceleration methods, 10 eigenmodes were used to approximate the direct solution. For the Craig–Bampton method, 10 eigenmodes were retained for each material substructure. From the FRF, it can be observed that several resonance peaks associated with the damped eigenmodes decrease in amplitude, which indicates energy dissipation caused by damping. Moreover, the influence of damping becomes more pronounced at higher frequencies, showing that stiffness-proportional damping contributes more significantly than mass-proportional damping. Despite these effects, damping does not alter the eigenmodes themselves, as the mode shapes remain unchanged between the undamped and damped cases for all eigenfrequencies.

As shown in Figure 2, the first mode shape primarily causes longitudinal deformation of the snowboard beneath the rider's feet, resembling a half-sine wave. The second mode shape produces transverse deformation similar to a half-cosine wave, which may introduce stresses in the snowboard bindings. The third mode shape results in anti-phase motion of the feet, which could influence the rider's skiing style. The pronounced resonance peaks at low frequencies indicate that the damping region has little influence in this range and is therefore ineffective for carving at low speeds. At higher modes, more complex deformation patterns appear, characterized by combinations of transverse and

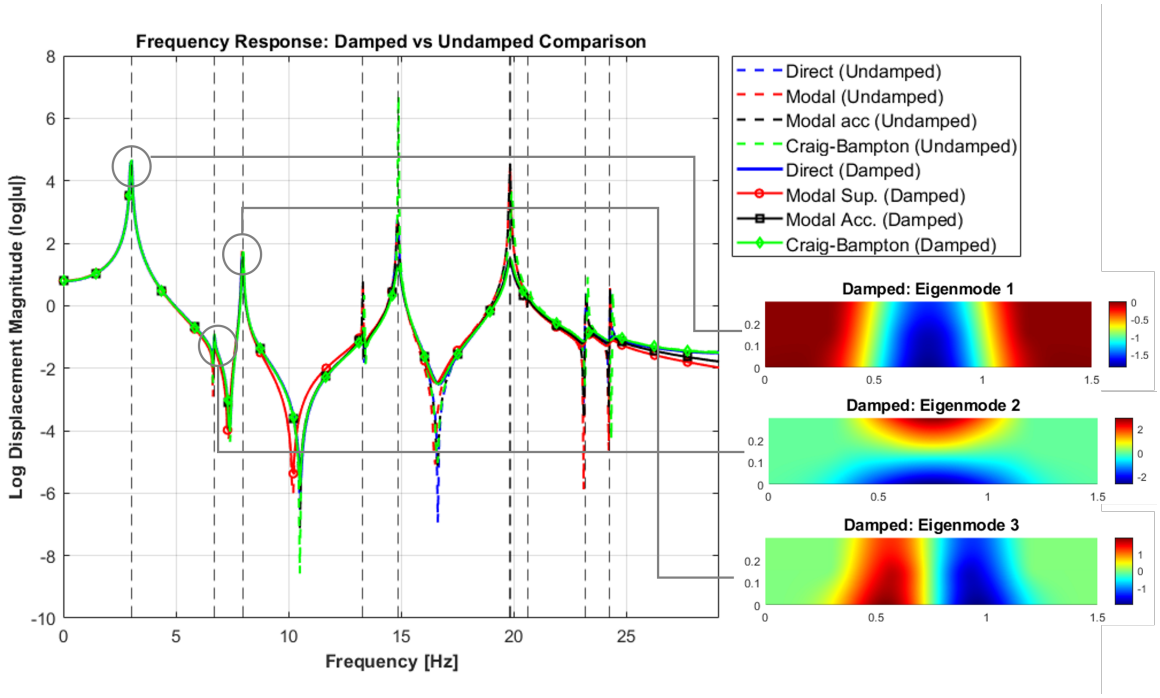


Figure 2: Frequency response function for undamped and damped case

longitudinal wave behaviour. However, at higher riding speeds, the snowboard is excited at higher frequencies where damping becomes more effective. As a result, the snowboarder experiences minimal deformation, meaning the riding style is unlikely to be significantly affected.

Table 2 compares the different solution methods in terms of computational time and mean square error relative to the direct solver. Although the direct solver requires the greatest computational effort, it also yields the most accurate results. For the undamped case, all reduced methods show relatively large errors compared to the direct solution, with only minor differences between modal superposition, modal acceleration, and the Craig–Bampton method. While modal acceleration provides a slight improvement in accuracy compared to conventional modal superposition, the improvement is marginal. The Craig–Bampton method remains the fastest approach, although this comes at the cost of a slightly higher error. For the damped case, modal acceleration is computationally more expensive than conventional modal superposition but provides results that more closely approximate the direct solution. The Craig–Bampton method offers the best balance between efficiency and accuracy, as it is the fastest reduced method while maintaining high accuracy, making it the preferred choice for engineering applications. The significant reduction in error for the Craig–Bampton method in the damped system further highlights its robustness when damping effects are included.

Table 2: Undamped and damped solver comparison

Undamped	Direct solve	Model sup.	Model acc.	Craig-Bampton
Computing time	27.6186	10.0201	6.4769	2.3017
Mean square error [%]	-	139.94	140.04	141.27
Damped	Direct solve	Model sup.	Model acc.	Craig-Bampton
Computing time	43.5504	4.1632	4.9352	3.9654
Mean square error [%]	-	95.90	95.98	0.0133

2.1 Energies of The System

Since the undamped case does not involve dissipation, the energy analysis focuses on the damped model. As the first natural frequency exhibits the highest excitation amplitude in the frequency response (Figure 2), the corresponding modal energies are of primary interest.

Figure 3 presents the spatial distributions of kinetic, potential, and dissipated energy for the first mode. The kinetic energy is concentrated in the center of the board, consistent with the large transverse velocities characteristic of the center of Mode 1. This localization reflects the mode shape geometry, in which the midspan region has the greatest displacement amplitude. In contrast, the potential energy is concentrated near the board's boundary condition, where the bending strain is highest, peaking at the inner corner of the boundary conditions.

The dissipated energy per excitation period is spatially confined to the damping section, as expected from its localized material placement. A clear correlation between the kinetic and dissipated energy distributions is observed, since both depend on the local velocity field. The difference in their magnitudes, however, is governed by the damping matrix, which determines the proportion of kinetic energy dissipated per cycle. This spatial correspondence confirms that the damping elements are most effective when placed in regions of high modal velocity, providing a basis for optimizing damping section placement.

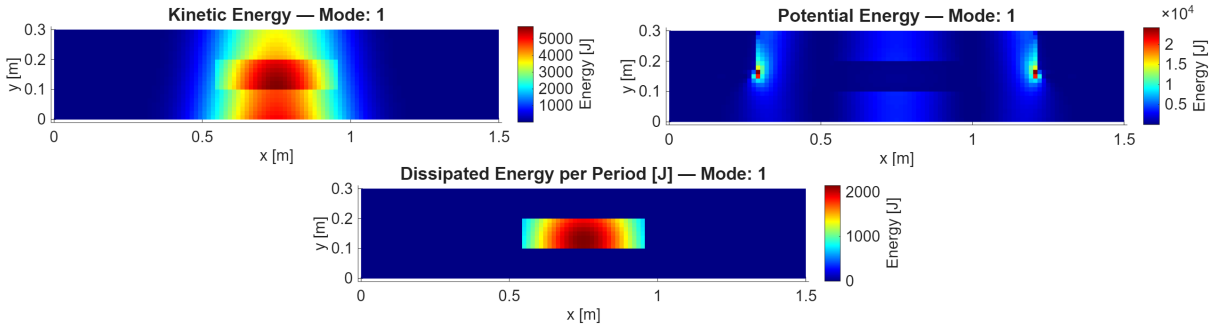


Figure 3: Kinetic, potential and dissipated energies of the snowboard for modeshape 1

To further understand the role of the damping element in the modal energy distribution, the structural power flow is examined. Figure 4 illustrates the structural power flow (Poynting vector field) for the first mode, quantifying the directional transport of vibrational energy through the board.

The outward orientation of the vectors at the damping section indicates that the total energy within it exceeds the locally dissipated energy. Although the vector magnitudes are reduced inside the layer, the net energy flux remains directed outward, meaning the damping element does not dissipate sufficient energy to reverse the flow direction.

For the Poynting vectors to point inward toward the damping section, one of two conditions must be met, either the damping coefficient is increased until the dissipated energy exceeds the local kinetic energy, or the mass of the damping element is reduced, by lowering its density ρ or thickness h, so that its contribution to the kinetic energy is smaller than that of the surrounding structure. In this configuration, the damping coefficient is too low and the density of the damping layer is too high, resulting in a local kinetic energy that exceeds the dissipated energy. Consequently, the damping section does not dissipate sufficient vibrational energy to establish a net inward power flux, as would be expected of an effective global energy sink.

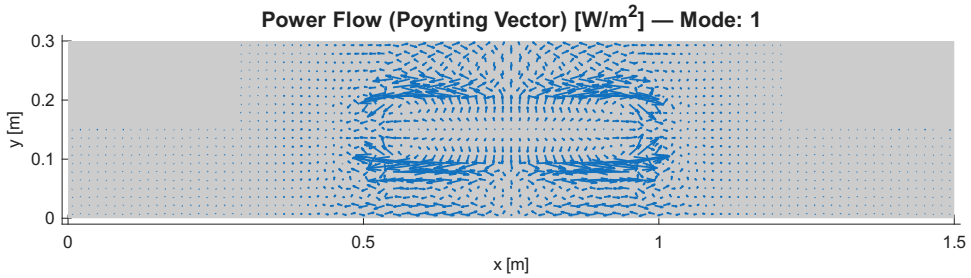


Figure 4: Poynting vector plot of the power flow in the snowboard

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References

[1] N. Aage and J. S. Jensen, Lecture Notes: 41813 FEM for sound and vibrations. 2025.